THERMO SENSE – POWER BUDGETING ON A CHIP

Capstone Report | PDEs, Transforms & Optimization Techniques

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Abstract

This project investigates how a fixed 5 W power budget can be distributed among four internal heat-source regions to reduce the maximum steady-state temperature on a chip. The workflow connects Fourier's law and energy conservation to the elliptic Poisson equation, finite-difference discretization, sparse linear solving, and linear programming (LP). The PowerPoint reports a uniform-allocation peak of 352.91 K and an optimized peak of 342.01 K, a reduction reported as approximately 10.89 K while using the full 5 W budget. These results depend on the stated geometry, source regions, conductivity, and boundary assumptions.

1. Field Model

The chip is modelled as a two-dimensional die with internal heat generation. Fourier's law relates heat flux to the temperature gradient:

$q = -k\nabla T$

Here q is heat flux, k is thermal conductivity, and T is temperature. At steady state, energy conservation relates the divergence of heat flux to the internal source term Q. For constant conductivity, the governing equation is $-k\nabla^2 T = Q$, or $\nabla^2 T = -Q/k$. Heat therefore flows from higher to lower temperature.

2. PDE Formulation

The steady two-dimensional heat-conduction model is the Poisson equation:

$\partial^2 T/\partial x^2 + \partial^2 T/\partial y^2 = -Q/k$

This is an elliptic PDE. Dirichlet boundary conditions fix all four edges at 300 K. The presentation specifies a 10 × 10 mm die, a 40 × 40 grid, conductivity k = 150 W/(m·K), four symmetric 4 × 4-cell source regions, and an assumed thickness of 1 mm. The model is steady-state; transient heat capacity is not included.

3. Transform Solution and Physical Interpretation

Thermal impedance is defined as $Z_{th}(s) = \Delta T(s)/P(s)$, relating temperature rise to applied power in the transform domain. The steady-state interpretation corresponds to $s \to 0$. According to the presentation, the notebook solves the steady-state PDE directly rather than implementing a transient transform-domain solver. Thermal impedance therefore provides the physical interpretation of the power-to-temperature relationship within the project pipeline.

4. Discretization

The finite-difference method (FDM) approximates the Laplacian with a five-point stencil at each interior grid point:

$\nabla^2 T(i,j) \approx [T(i+1,j)+T(i-1,j)+T(i,j+1)+T(i,j-1)-4T(i,j)] / h^2$

Applying the stencil and incorporating fixed boundary values gives a sparse algebraic system. The presentation describes the system as $A\theta = Q/k$ and identifies SciPy's spsolve as the solver. Unit-watt responses from the individual source regions form a thermal response matrix H, producing the linear model $T = T_{amb} + Hq$. This response matrix is used by the optimization stage.

5. LP Formulation

The objective is to choose block powers q■–q■ to minimize the hottest predicted grid temperature. An auxiliary variable t represents the peak-temperature bound:

Minimize: t

Subject to: T■(q) ≤ t for every grid point i

$q_■ \geq 0$ for j = 1, 2, 3, 4

$q_■ + q_■ + q_■ + q_■ = 5$ W

Because temperature is linear in the source powers under the model assumptions, the optimization is a linear program. The presentation identifies scipy.optimize.linprog as the solver. The total-power equality ensures that the uniform and optimized cases both use the full 5 W budget.

6. Results and Verification

The presentation reports the following comparison:

Measure	Uniform allocation	Optimized allocation
Block powers (W)	[1.25, 1.25, 1.25, 1.25]	[0.801, 1.364, 1.417, 1.417]
Total power (W)	5.000	≈ 4.999 (rounded values)
Peak temperature (K)	352.91	342.01
Peak reduction	—	≈ 10.89 K (reported)

The stated verification checks are: (1) confirm LP solver success; (2) check that all optimized powers are non-negative; (3) check that the total allocation equals 5 W; and (4) compare the optimized peak with the uniform baseline. The reported peak falls from 352.91 K to 342.01 K. The displayed optimized powers sum to 4.999 W due to rounding, so full-precision solver values should be used for the exact budget check.

The presentation notes that 343 K and 327 K from the earlier brief are reference values, not outputs generated by this simulation.

7. Pipeline Reflection

The project connects the course topics into a single workflow: vector calculus and Fourier's law establish the field model; the PDE describes steady conduction; thermal impedance interprets the relation between power and temperature rise; finite differences convert the PDE to a sparse algebraic system; and LP selects a feasible allocation that minimizes the peak temperature. The implementation stage joins these components and checks solver status, power constraints, and peak-temperature improvement.

Limitations include fixed-temperature edges, constant conductivity, idealized source footprints, a finite 40 × 40 grid, and steady-state-only behaviour. Future improvements include grid-convergence testing, realistic cooling boundaries, and transient thermal modelling. Keep the assumed 1 mm thickness consistent across the report, slides, and code.

Pipeline Table (required in Section 7)

Owner initials are based on the team-member names shown in the presentation.

Stage	Math Concept	Course	Equation / Implementation	Owner
1. Field Model	Vector calculus; conservation	PDEs	$q = -k\nabla T$; $-k\nabla^2 T = Q$	MY
2. PDE	Elliptic Poisson equation; BCs	PDEs	$\nabla^2 T = -Q/k$; edge $T = 300$ K	MY
3. Transform	Thermal impedance; steady limit	Transforms	$Z_{th}(s) = \Delta T(s)/P(s)$; $s \to 0$	SV
4. Discrete	Finite differences; sparse solve	Numerical methods	5-point stencil; $A\theta = Q/k$; spsolve	PMS
5. LPP	Linear programming; epigraph variable	Optimization	min t; $T_■ \le t$; $q_■ \ge 0$; $\Sigma q_■ = 5$ W	DSK
6. Verification	Numerical and constraint checks	Python implementation	Solver success; budget and peak comparison	BCR

8. References + Repository

• Capstone Project S3 EC, Problem A brief.

• Incropera et al., Fundamentals of Heat and Mass Transfer, 7th ed., Wiley, 2011.

• S. V. Patankar, Numerical Heat Transfer and Fluid Flow, 1980.

• S. Boyd and L. Vandenberghe, Convex Optimization, Cambridge University Press, 2004.

• SciPy documentation: sparse matrices, spsolve, and scipy.optimize.linprog.

• NumPy documentation and Matplotlib documentation.

Repository / Colab URL: Not visible in the supplied files. Add the team's shareable Colab or Git repository link before submission.